

AKHIL & KARTHIKA DIGITAL WEDDING INVITATION

Complete Project Files, Architecture & Code Usage Documentation

React 18 · Vite · Tailwind CSS · Framer Motion · Lucide React · HTML5 Audio

Wedding Date: 24 January 2027 · **Venue:** Prince Convention Centre, Alappuzha, Kerala

Architecture: Frontend-only Single-Page Interactive Web Application

1. Technical Architecture & Design System

This project is a high-aesthetic, production-ready, interactive digital wedding invitation designed as a single-page progressive journey. It follows editorial typography, warm Kerala-inspired aesthetic tones, delicate hand-drawn SVG botanicals, organic blob frames, and fluid Framer Motion choreography. The application is frontend-only, requiring no server or database infrastructure for deployment.

Category	Token / Value	Visual Purpose
Surfaces	#FAF7F2 (Ivory) / #F5EFE6 (Cream)	Warm textured background & soft modal cards
Brand Accent	#7A1C28 (Maroon) / #8B2635 (Terracotta)	Headings, primary CTA buttons & active highlights
Soft Accent	#E8D3D3 (Blush) / #7D8B75 (Sage Olive)	Borders, card dividers & botanical leaf elements
Typography	Playfair Display, Plus Jakarta Sans, Caveat	Display serifs, legible UI body, & handwritten notes
Motion Engine	Framer Motion 11.x & CSS Keyframes	Mask reveals, scroll triggers, micro-interactions & floating particles

2. Project Directory & File Map

```
d:\Antigravity Projects\Invitation\ ■■■ index.html # Root HTML entry with fonts & CDN preconnects ■■■  
package.json # Project metadata, dependencies & scripts ■■■ vite.config.js # Vite bundler & React  
plugin config ■■■ tailwind.config.js # Theme tokens, custom fonts, shadows & animations ■■■  
postcss.config.js # PostCSS Tailwind & Autoprefixer plugin config ■■■ src/ ■■■ main.jsx # React 18  
DOM mount point ■■■ App.jsx # State orchestrator & section flow coordinator ■■■ index.css # Custom  
utilities, washi-tape styles, scrollbars ■■■ data/ ■■■ weddingData.js # Centralized event data,  
images, audio & copy ■■■ components/ ■■■ OpeningScreen.jsx # Fullscreen curtain reveal with "OPEN  
INVITATION" CTA ■■■ Navbar.jsx # Sticky glassmorphic header with smooth scroll & mobile drawer ■■■  
Hero.jsx # Asymmetric editorial grid with organic curved photo frame ■■■ Countdown.jsx # Live dynamic  
countdown timer to Jan 24, 2027 11:00 AM IST ■■■ WeddingDetails.jsx # Ceremony details, Google Maps  
link, Add-to-Calendar & polaroid ■■■ Gallery.jsx # Staggered asymmetric photo collage with rotations &  
tape ■■■ Lightbox.jsx # Fullscreen interactive photo modal with keyboard Escape ■■■ QuoteSection.jsx  
# Typographic love quote breathing space with drifting botanicals ■■■ RSVP.jsx # Accessible attendance  
form with validation & confetti state ■■■ ThankYou.jsx # Panoramic sunset silhouette banner with  
maroon gradient ■■■ MusicControl.jsx # Floating sound player with animated equalizer bars ■■■  
DecorativeElements.jsx # Reusable hand-drawn SVG botanicals, tape & monogram
```

3. Detailed File & Source Code Breakdown

FILE: index.html	Role: Root HTML Document Entry Point
Purpose: Provides the HTML container (<code><html></code>), loads external web typography, sets responsive viewport configurations, and declares browser performance optimizations. • Preconnect Links: Establishes early TLS handshakes with <code>fonts.googleapis.com</code>, <code>fonts.gstatic.com</code>, and <code>images.unsplash.com</code> to eliminate render-blocking network latency. • Google Web Fonts: Imports <code>Playfair Display</code>, <code>Cormorant Garamond</code>, <code>Plus Jakarta Sans</code>, and <code>Caveat</code>. • Meta Information: Includes theme-color <code>#FAF7F2</code>, wedding open-graph descriptions, and rings favicon.	
FILE: package.json	Role: Project Manifest, Dependencies & Scripts
Purpose: Specifies project metadata, runtime packages (React 18, Framer Motion, Lucide React, Canvas Confetti), build tools (Vite 5, Tailwind CSS, PostCSS, Autoprefixer), and terminal commands. • Runtime Dependencies: <code>react</code>, <code>react-dom</code>, <code>framer-motion</code> (for choreographed reveals and layout animations), <code>lucide-react</code> (clean modern icons), <code>canvas-confetti</code> (for RSVP celebration). • Scripts: <code>npm run dev</code> (starts Vite dev server at port 3000), <code>npm run build</code> (creates production distribution bundle in <code>/dist</code>), <code>npm run preview</code> (serves production build).	
FILE: vite.config.js & postcss.config.js	Role: Build Bundler & CSS Tooling Pipeline
Purpose: Configures Vite's lightning-fast JSX transform with <code>@vitejs/plugin-react</code> and wires PostCSS to execute Tailwind CSS and vendor autoprefixing. • Vite Server Port: Standardized to port 3000 with HMR enabled. • PostCSS Setup: Enables Tailwind utility compilation and vendor-prefix injection for broad cross-browser compatibility.	
FILE: tailwind.config.js	Role: Design System Token & Theme Engine
Purpose: Extends the default Tailwind palette with wedding color tokens, custom serif/script font families, organic blob border radiuses, warm drop shadows, and subtle floating keyframe animations. • Color Tokens: <code>ivory</code>, <code>maroon</code>, <code>terracotta</code>, <code>blush</code>, <code>sage</code>, <code>gold</code>, and <code>charcoal</code>. • Custom Shadows: <code>shadow-polaroid</code>, <code>shadow-card</code>, and <code>shadow-maroon</code> for tactile depth. • Keyframe Animations: <code>float-slow</code> (drifting botanical motion) and <code>pulse-subtle</code>.	

FILE: src/index.css	Role: Global Stylesheet & Custom CSS Utilities
Purpose: Imports Tailwind layers, configures custom warm-ivory scrollbars, creates washi-tape photo edges, and enforces strict `@media (prefers-reduced-motion)` accessibility standards.	
• .tape-sticker: Realistic translucent washi-tape overlay with dashed gold borders and subtle blur. • .blob-shape-1 / .blob-shape-arch: Organic curved clip profiles for asymmetric photo presentation. • Accessibility: Automatically pauses or minimizes animations when users have reduced motion enabled in their OS.	
FILE: src/data/weddingData.js	Role: Centralized Content, Event Metadata & Asset Strategy
Purpose: Decouples copy, dates, coordinates, and images from UI presentation. Serves as the single source of truth for all components across the site.	
• Event ISO Timestamp: `2027-01-24T11:00:00+05:30` consumed by the Countdown engine. • Venue Data: Address, coordinates, and direct Google Maps search query link for Alappuzha, Kerala. • Editorial Photography Array: 6 curated Unsplash photos with custom aspect ratios, rotation angles, alt texts, and handwritten captions.	
FILE: src/App.jsx	Role: Application Orchestrator & State Coordinator
Purpose: Controls the lifecycle of the entire experience: manages the opening reveal state, initializes and controls background audio with HTML5 Audio API, toggles body scroll locking, and renders all 8 sections sequentially.	
• Scroll Lock Effect: Enforces `document.body.style.overflow = 'hidden'` until the user clicks 'OPEN INVITATION'. • Audio API Controller: Plays `wedding-ambient.mp3` on the user's explicit interaction (avoiding browser autoplay blocks) with smooth play/pause state synchronization. • Component Pipeline: Renders OpeningScreen -> Navbar -> Hero -> Countdown -> WeddingDetails -> Gallery -> QuoteSection -> RSVP -> ThankYou -> MusicControl.	

FILE: src/components/OpeningScreen.jsx	Role: Section 01: Fullscreen Invitation Reveal Overlay
Purpose: Presents an immersive digital envelope / card reveal. Displays family blessings, couple names, wedding date, and the primary 'OPEN INVITATION →' button. • Framer Motion Curtain Reveal: Uses `` with `y: '-100%` and smooth cubic-bezier curve to elegantly reveal the main invitation. • Framed Couple Portrait: Features couple photo with handwritten 'Better Together' note and floating botanical doodles. • Accessibility: Equipped with `role='dialog`, `aria-modal='true`, and keyboard-navigable button.	
FILE: src/components/Navbar.jsx	Role: Sticky Navigation Header & Section Indicator
Purpose: Sticky glassmorphic bar that appears after opening. Provides rapid navigation to sections with smooth scrolling, active section tracking, and mobile drawer menu. • Active Scroll Spy: Computes window scroll position and highlights current section with an animated Framer Motion layout indicator. • Mobile Drawer: Collapsible mobile menu with hamburger toggle for touch screens. • Quick RSVP CTA: High-contrast maroon button linking directly to the RSVP form.	
FILE: src/components/Hero.jsx	Role: Section 02: Asymmetric Editorial Hero
Purpose: The primary visual anchor of the site. Displays handwritten 'Together forever', grand serif names, event highlight badges (Date, Time, Venue), and an organic curved picture frame. • LCP Image Optimization: Primary portrait tagged with `fetchpriority='high` for instantaneous rendering. • Event Summary Cards: Quick-glance 3-column badge grid displaying Makaram date, Muhurtham time, and Alappuzha location. • Interactive Actions: 'Our Journey ↓' smooth scroll button and 'Reserve Attendance' link.	

FILE: <code>src/components/Countdown.jsx</code>	Role: Section 03: Live Real-Time Countdown Engine
Purpose: Dynamic live countdown calculating the remaining time down to the exact second until January 24, 2027 11:00 AM IST.	
• JavaScript Time Engine: <code>setInterval</code> hook calculating Days, Hours, Minutes, and Seconds from <code>Date.now()</code> to target ISO timestamp. • Delicate Cream Panels: 4-column responsive grid with thin maroon borders and subtle hover lift. • Floral Flankers: Symmetrical hand-drawn SVG botanical ornaments framing the title.	

FILE: <code>src/components/WeddingDetails.jsx</code>	Role: Section 04: Ceremony Specifications & Venue Card
Purpose: Provides guests with ceremony timings, Google Maps directions, Add-to-Calendar functionality, and a taped polaroid of the Kerala destination.	
• Google Calendar Generator: Encodes event title, dates (<code>20270124T053000Z</code>), venue, and Sadya feast details into a direct calendar link. • Polaroid Card: Styled with washi-tape sticker, Kerala backwaters photograph (<code>loading="lazy"</code>), and handwritten 'See you there! 🍷' script. • Map Direction CTA: Direct link opening Google Maps for Prince Convention Centre, Alappuzha.	

FILE: <code>src/components/Gallery.jsx & Lightbox.jsx</code>	Role: Section 05: Asymmetric Photo Collage & Interactive Lightbox
Purpose: Presents pre-wedding editorial photos in a staggered, asymmetric layout with rotational angles, washi tape, and click-to-zoom modal.	
• Asymmetric Stagger: Alternating rotations (<code>-rotate-2</code>, <code>rotate-2</code>) and column spans (<code>col-span-7</code>, <code>col-span-5</code>) creating an editorial magazine feel. • Lightbox Modal: Accessible dialog with backdrop blur, full-resolution image display, photo caption, and keyboard <code>Escape</code> listener. • Performance: Below-the-fold images use <code>loading="lazy"</code> and <code>decoding="async"</code>.	

FILE: <code>src/components/QuoteSection.jsx</code>	Role: Section 06: Typographic Love Pause
Purpose: Provides visual breathing space with large quotation marks around 'Two hearts, one beautiful beginning.' flanked by drifting SVG leaf vectors.	
• Grand Serif Quote: 60px display quotation marks with gold divider line and pulsing heart icon. • Restrained Whitespace: High-contrast, minimal layout encouraging a thoughtful emotional pause.	
FILE: <code>src/components/RSVP.jsx</code>	Role: Section 07: Interactive Attendance Form & Celebration Card
Purpose: Accessible attendance submission form with real-time validation, custom radio buttons, no guest-count field, and celebratory confetti state.	
• Form State & Validation: Verifies guest name and ensures attendance selection ('Joyfully Accept' vs. 'Regretfully Decline'). • Celebratory Confetti: Integrates `canvas-confetti` with maroon and gold color scheme upon affirmative RSVP. • Confirmation View: Smoothly replaces form with personalized thank-you card and allows resetting or editing response.	
FILE: <code>src/components/ThankYou.jsx</code>	Role: Section 08: Full-Width Cinematic Sunset Banner
Purpose: The grand finale of the invitation: a full-width sunset silhouette of the couple overlayed with a deep maroon gradient filter and golden typography.	
• Cinematic Backdrop: High-resolution sunset image with multi-stop dark maroon and charcoal gradient overlay. • Grand Typography: 'THANK YOU FOR BEING PART OF OUR SPECIAL DAY', monogram seal, and wedding date. • Footer Metadata: Copyright, blessings note, and handcrafted love credit.	

FILE: <code>src/components/MusicControl.jsx</code>	Role: Floating Background Audio Controller
Purpose: Unobtrusive bottom-right audio widget that displays sound equalizer animations when playing and toggles audio on demand. • Equalizer Animation: 3 pulsing bar bars simulating live acoustic frequency waves. • Status Dot: Emerald pulsing indicator showing active audio stream. • Accessible Controls: Keyboard focusable, ARIA title, and smooth Framer Motion entry.	
FILE: <code>src/components/DecorativeElements.jsx</code>	Role: Reusable SVG Botanical Art & Visual Accents
Purpose: Provides vector botanical branches, floral line-art ornaments, doodle hearts, sparkles, washi tape, and couple monogram badges. • Pure SVG: Zero raster asset overhead; lightweight, scalable, and crisp on all retina screens. • Washi Tape: Polygon clip-path creating realistic ripped paper edges.	

4. Execution Commands & Verification Summary

To launch and run the project locally via Command Prompt:

```
cd /d "d:\Antigravity Projects\Invitation"  
npm run dev (Runs development server at http://localhost:3000)
```

To compile and test the optimized production build:

```
npm run build (Compiles Vite production bundle)  
npm run preview (Serves preview at http://localhost:4173)
```

Validation Criteria	Implementation Result	Status
Opening Curtain Mask Reveal	Framer Motion smooth mask unroll with audio trigger	PASS
Live Real-Time Countdown	Dynamic countdown to 24 January 2027 11:00 AM IST	PASS
Wedding Venue & Maps	Prince Convention Centre, Alappuzha + Google Maps & Calendar	PASS
Asymmetric Editorial Gallery	Staggered layout, tape stickers, rotations & full Lightbox modal	PASS
Frontend RSVP Form	Name validation, attendance radio, no guest count, confetti	PASS
Audio Controller	Floating soundwave bars, play/pause toggle, acoustic serenade	PASS
Accessibility & Responsiveness	ARIA modal, focus management, reduced-motion, mobile drawer	PASS
Production Build	Vite 5 bundle compiled cleanly with 0 errors / 0 warnings	PASS